

# STEPHEN BOILEAU

## Technical Artist & Developer | Unity & Unreal Engine | Real-Time 3D Pipelines

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## PROFESSIONAL SUMMARY

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Technical Artist and Developer with 10+ years of experience delivering real-time and pre-rendered interactive experiences for themed entertainment, robotics, VR/AR, and immersive media. Strong background in Unity (C#), Unreal Engine (C++/Blueprints), Python tooling, real-time character systems, rendering pipelines, asset optimization, and cross-platform deployment across Android, iOS, PC, and Meta Quest. Proven ability to bridge art, engineering, design, and creative direction while building reliable, artist-friendly technical solutions across the full development lifecycle.

## CORE SKILLS

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- **Engines & Development:** Unity, Unity 6, C#, Unreal Engine, C++, Blueprints, gameplay systems, real-time application development
- **Technical Art & Pipelines:** pipeline creation, asset exporting/importing, asset validation, asset optimization, Autodesk Maya tools, Python, PyMEL, batch automation, review and delivery workflows
- **Rendering & Performance:** rendering pipelines, Movie Render Queue, high-resolution and large-format capture, custom render dimensions, 360 stereo rendering (OWL / Off World Live), MetaHuman, performance profiling, mobile/VR optimization
- **Interactive 3D / VR / AR:** Meta Quest, Oculus, Android, iOS, PC, VR development, AR development, immersive media, TouchDesigner, OpenCV
- **UI / UX Implementation:** in-engine tools, operator-facing production interfaces, mobile game controls, non-technical user workflows
- **Networking & Systems:** Unity Netcode for GameObjects, multiplayer systems, client-server fundamentals, production debugging
- **Source Control & Collaboration:** Perforce, P4 server administration, Git, Plastic SCM, Jira, AI-assisted workflows, cross-functional collaboration

## PROFESSIONAL EXPERIENCE

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### Super 78 — Technical Artist & Developer

*Los Angeles, CA | December 2021 – Present*

- Develop real-time interactive applications and production pipelines using Unity C# and Unreal Engine C++/Blueprints for themed entertainment attractions, VR deliverables, and live character systems.
- Build and maintain Unreal Engine rendering pipelines, including Movie Render Queue workflows, high-resolution and large-format capture at unique render dimensions, and 360 stereo rendering using OWL / Off World Live.
- Design Python-based automation for batch rendering, video conversion for review, Slack notification workflows, and standardized client-ready media delivery.
- Create custom Python, PyMEL, and Maya tools to automate asset preparation, exporting/importing, validation, naming, and production support tasks.
- Integrate MetaHuman workflows, including animation and facial mocap cleanup, for high-fidelity real-time and rendered character output.
- Develop operator-facing UI/UX tools inside Unreal Engine for live show control by performers and non-technical production users.
- Serve as Perforce (P4) server administrator across multiple teams and contributors; also maintain Git and Plastic SCM workflows. Optimize projects for real-time performance across VR, Android, Meta Quest, PC, and large-format media, and troubleshoot renders, Blueprints, and legacy systems under production deadlines.

## **Embodied Inc. — Technical Artist**

*Pasadena, CA | June 2018 – December 2021*

- Helped bring Moxie, a real-time interactive social robot, to market — contributing character rigging, animation preparation, asset integration, C# development, and production pipeline tooling.
- Developed Python automation tools to improve asset processing, validation, workflow consistency, and iteration speed.
- Improved asset stability, documentation, and mobile graphics performance for a consumer-facing real-time character platform.
- Collaborated across engineering, animation, and design within Git and Jira-based production workflows.

## **DeepBlue Worlds — Junior Technical Artist**

*Los Angeles, CA | February 2014 – June 2018*

- Contributed to Unity-based interactive applications and games using C#, Autodesk Maya, and real-time engine workflows.
- Supported gameplay integration, asset preparation, rigging, and animation for Android and iOS consumer applications, including AR and interactive educational experiences.
- Managed Maya-based asset workflows and Git-based project organization for real-time deployment.

## **Mt. San Antonio College — Lab Technician / Teacher Assistant**

*Walnut, CA | August 2013 – December 2016*

- Assisted instruction in Autodesk Maya, introductory modeling, animation, and game-development coursework; provided technical guidance on 3D workflows and production tools.

## **SELECTED PROJECTS**

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### **NightFlight Expedition — Dollywood (2026, opening soon)**

- Technical Artist and Developer on Unreal Engine media production for the world's first indoor family hybrid coaster and whitewater river raft ride — built large-format rendering pipelines at unique, non-standard render dimensions for the attraction's projection systems, including Movie Render Queue setup, render troubleshooting, and Perforce administration.

### **Sasquatch Mountain Opening — Unreal Engine Media Pipeline**

- Built and supported Unreal Engine rendering workflows for high-resolution output and content delivery — MRQ pipelines, automated review rendering with Slack notifications, MetaHuman animation and mocap cleanup, 360 stereo rendering (OWL plugin), and P4 server administration across multiple teams.

### **Live Character System — Unreal Engine (2024)**

- Built and maintained a real-time animation system supporting hundreds of animation states for a live character performing daily in a major theme park attraction, with a custom operator-friendly GUI enabling performers to control the character live.

### **Chimelong Spaceship — Live Unity Character Experience (2022)**

- Joined late-stage development and resolved major performance issues — optimizing Maya assets and Unity code to raise runtime performance from roughly 13 FPS to above 60 FPS without sacrificing visual quality.

### **Client Theater Visualizer — VR Application (2023)**

- Designed and built a Meta Quest VR tool letting clients review in-progress media inside accurately scaled virtual theaters, with to-scale Maya models, precise UV mapping for curved projection surfaces, and a simple in-headset media loading workflow.

### **Princess Cruises — 360 Dining Experience (2023)**

- Developed TouchDesigner and Python systems using OpenCV to track guest motion and gestures, driving real-time interactive graphics for an immersive theatrical dining environment.

### **Unity Multiplayer Strategy Game — Personal Project (in closed beta)**

- Designed and developed an online multiplayer turn-based strategy game in Unity 6 for Android and iOS using Unity Netcode for GameObjects, with mobile-focused UI/UX, touch controls, and client-server gameplay systems.

### **Dino on My Desk AR (2016)**

- Sole developer of an educational AR app for Android and iOS built in Unity with C#, managing gameplay, platform support, Maya asset workflows, and Git-based project organization.

## **EDUCATION**

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Mt. San Antonio College — Coursework in Animation, Game Development, and Autodesk Maya